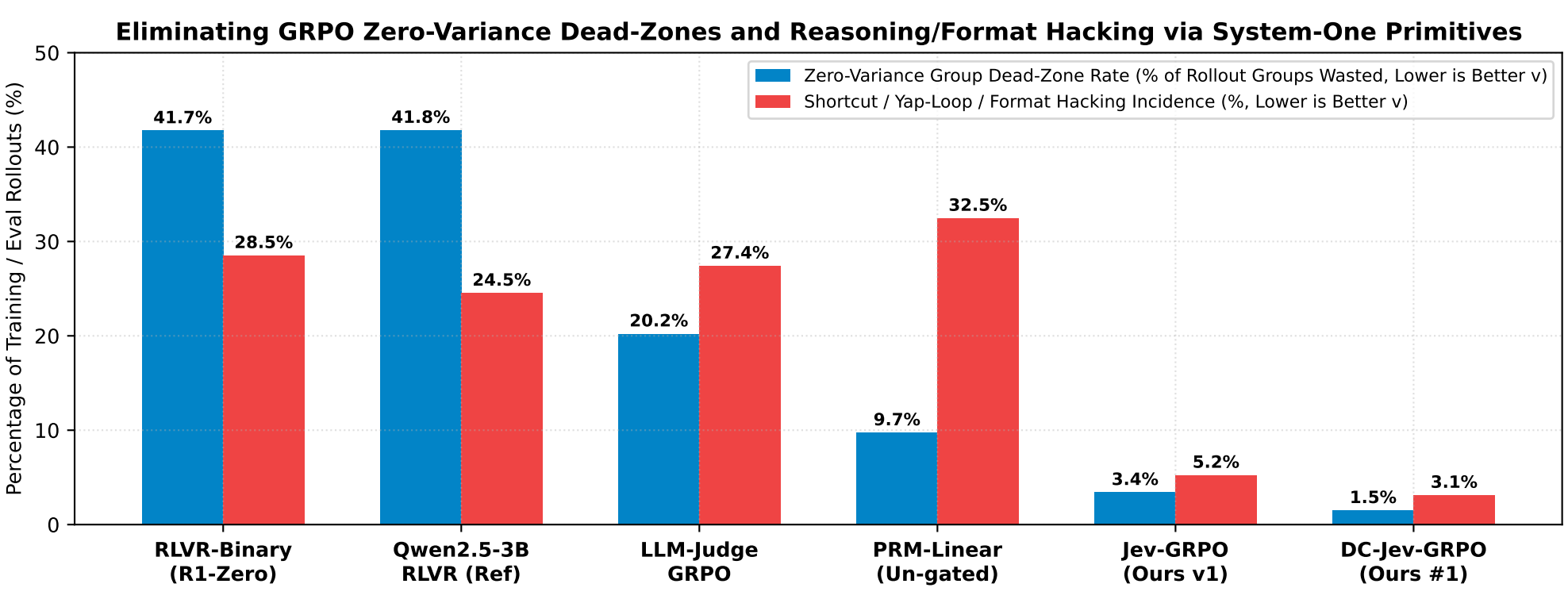
Eliminating GRPO Zero-Variance Dead-Zones and Reasoning/Format Hacking via System-One Primitives
Zero-Variance Group Dead-Zone Rate (% of Rollout Groups Wasted, Lower is Better v)
Shortcut / Yap-Loop / Format Hacking Incidence (%, Lower is Better v)
Percentage of Training / Eval Rollouts (%)
0
10
20
30
40
50
41.7%
28.5%
41.8%
24.5%
20.2%
27.4%
9.7%
32.5%
3.4%
5.2%
1.5%
3.1%
RLVR-Binary (R1-Zero)
Qwen2.5-3B RLVR (Ref)
LLM-Judge GRPO
PRM-Linear (Un-gated)
Jev-GRPO (Ours v1)
DC-Jev-GRPO (Ours #1)